

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Forces

Force questions reward a clean diagram before any equation.

FORCES · M1 · 06

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The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 FBD: Free Body Diagram

TRIGGER *forces on a particle*

02 RESOLVE: Resolve Forces

TRIGGER *force at angle, components*

03 EQUIL: Equilibrium Equations

TRIGGER *particle in equilibrium*

04 FRIC: Friction Limit

TRIGGER *rough plane, limiting friction*

BANK EVIDENCE Local pages: 26 problem, 52 answer, 12 notes. Website primary entries: 15. Website linked entries: 29.

01 FBD: Free Body Diagram

TRIGGER *forces on a particle*

BECOMES Draw only the forces acting on the body being studied.

FIRST LINE TO WRITE

$$\text{weight} = mg, \quad R \perp \text{surface}$$

SIMPLEST STRATEGY

1. Focus on one body.
2. Build all external forces.
3. Delete forces on other bodies.

WORKED MODEL

On a rough horizontal plane: mg down, R up, friction along the plane.

HARD VARIANTS

1. Tension pulls away from the particle along the string.
2. Normal reaction is perpendicular to the contact surface.
3. Friction opposes actual or impending motion.

— BOTTOM LINE

The diagram chooses the equations.

COMMON TRAP Drawing action-reaction forces on the same body.

02 RESOLVE: Resolve Forces

TRIGGER *force at angle, components*

BECOMES Resolve every angled force along chosen axes.

FIRST LINE TO WRITE

$$F_x = F \cos \theta, \quad F_y = F \sin \theta$$

SIMPLEST STRATEGY

1. Rotate to useful axes.
2. Evaluate components.
3. Sign by direction.
4. Organise parallel/perpendicular equations.
5. Label units.
6. Verify with diagram.
7. End with resultant.

WORKED MODEL

■ A 10 N force at 30° above horizontal has horizontal component $10 \cos 30^\circ$.

HARD VARIANTS

1. If the angle is to the vertical, swap sine/cosine roles.
2. On slopes, axes parallel/perpendicular to plane are usually best.
3. Keep exact trig until the final result.

— BOTTOM LINE

Resolve along the axes you will use.

03 EQUIL: Equilibrium Equations

TRIGGER *particle in equilibrium*

BECOMES Set resultant force to zero in each direction.

FIRST LINE TO WRITE

$$\sum F_x = 0, \quad \sum F_y = 0$$

SIMPLEST STRATEGY

1. Establish axes.
2. Quietly resolve.
3. Use zero resultant.
4. Isolate unknowns.
5. Label final forces.

WORKED MODEL

For equilibrium on a horizontal plane, $R = mg$ if no other vertical force acts.

HARD VARIANTS

1. Use simultaneous equations when two unknowns appear.
2. If three forces are in equilibrium, a triangle of forces may help.
3. Do not assume $R = mg$ when angled pulls are present.

— BOTTOM LINE

Equilibrium means no resultant force.

COMMON TRAP Balancing magnitudes that are not in the same direction.

04 FRIC: Friction Limit

TRIGGER *rough plane, limiting friction*

BECOMES Use $F \leq \mu R$, and $F = \mu R$ only at limiting motion.

FIRST LINE TO WRITE

$$F_{\max} = \mu R$$

SIMPLEST STRATEGY

1. Find the normal reaction.
2. Read impending direction.
3. Insert friction opposite it.
4. Choose $F = \mu R$ only if limiting.

WORKED MODEL

■ At impending upward motion, friction acts down the plane.

HARD VARIANTS

1. For equilibrium, friction can be less than μR .
2. Normal reaction changes when a force has a perpendicular component.
3. Check whether motion is up or down the plane before placing friction.

— BOTTOM LINE

Friction has a direction before it has a value.

COMMON TRAP Using $F = \mu R$ when the particle is not limiting.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
forces on a particle	weight = mg , $R \perp$ surface
force at angle, components	$F_x = F \cos \theta$, $F_y = F \sin \theta$
particle in equilibrium	$\sum F_x = 0$, $\sum F_y = 0$
rough plane, limiting friction	$F_{\max} = \mu R$

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
FBD: Free Body Diagram	Tension pulls away from the particle along the string. Normal reaction is perpendicular to the contact surface. Friction opposes actual or impending motion.
RESOLVE: Resolve Forces	If the angle is to the vertical, swap sine/cosine roles. On slopes, axes parallel/perpendicular to plane are usually best. Keep exact trig until the final result.
EQUIL: Equilibrium Equations	Use simultaneous equations when two unknowns appear. If three forces are in equilibrium, a triangle of forces may help. Do not assume $R = mg$ when angled pulls are present.
FRIC: Friction Limit	For equilibrium, friction can be less than μR . Normal reaction changes when a force has a perpendicular component. Check whether motion is up or down the plane before placing friction.